

Priyanka Chatterjee

Full-Stack Software Engineer · Data Pipelines & Insights APIs · Python · TypeScript · AWS

· 4+ Years

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SUMMARY

Full-stack software engineer with 4+ years building scalable data pipelines, insights APIs, and **distributed systems** in **Python**, **TypeScript**, and **Go** across **AWS**, Azure, and GCP. Demonstrated expertise in **operational excellence**—**monitoring**, **incident management**, performance optimization—and shipping full-lifecycle features from design through deployment. Engineered **Kafka**-based data pipelines processing 1,000+ events per second with dead-letter queues; optimized **PostgreSQL** queries powering compliance data analysis by 40% (cut processing time from 3 minutes to 1.5 minutes across 10,000+ records); led microservice migrations cutting resource-discovery latency 70–90% via **event-driven** architecture.

TECHNICAL SKILLS

Languages: Python, TypeScript, Go, JavaScript (ES6+), SQL, Bash, HCL

Backend & APIs: Microservices, REST APIs, gRPC, Event-Driven Architecture, Serverless, OAuth 2.0, Gin, Express.js, Node.js

Frontend: React, TypeScript, MUI, responsive design

Databases & Data: PostgreSQL, DuckDB, Parquet, SQL query optimization, query tuning, indexing, data pipelines, CTEs, jsonb

Messaging & Streaming: Apache Kafka, consumer groups, dead-letter queues (DLQs), event-driven pipelines, asynchronous processing, stream processing

Cloud & Distributed Systems: AWS (Lambda, EC2, RDS, S3, CloudWatch, X-Ray), Azure, GCP, scalability, latency optimization, throughput measurement, multi-cloud

DevOps & Observability: GitHub Actions, CI/CD pipelines, Docker, Kubernetes (working knowledge), Terraform, monitoring, logging, tracing, incident management, testing automation

Observability & Reliability: Datadog, OpenTelemetry (OTel), Logfire, performance monitoring, latency tracking, debugging, root cause analysis, operational excellence

Testing & Quality: Unit testing, integration testing, end-to-end testing, automated CI testing, code review, smoke tests

Core CS: Data structures and algorithms, distributed systems design, system architecture, API design, database design, performance optimization

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Engineered a **Kafka**-based, fire-and-forget **event-driven data pipeline** for policy-enforcement workflows with asynchronous consumer processing and a dead-letter queue (**DLQ**) for fault tolerance; ensured reliability and observability in high-volume data processing.
- Built core features for Tailpipe, a **Go**-based observability and data CLI on DuckDB + Parquet, lifting data-processing throughput 30% to 1,000+ records per second by optimizing chunking, batching, and streaming writes; reduced memory use to eliminate out-of-memory failures.
- Authored 11 **AWS** log-source data tables for Tailpipe in **Go** (CloudTrail, GuardDuty, SecurityHub, ALB/NLB/CLB/ELB access logs, S3 access, VPC flow, WAF, Network Firewall) with 25+ pre-built analytics/security detections; designed cross-platform smoke-test pipeline.
- Optimized complex **PostgreSQL** queries powering enterprise data analysis and compliance audits — rewriting IAM policy-evaluation queries with jsonb array operations, multi-stage CTEs, and conditional filtering — improving query performance 40% (cutting report-generation from 3 minutes to 1.5 minutes across 10,000+ records) and eliminating false-positive findings.
- Designed and developed a **REST API** microservice in **Go** using Gin framework with OAuth 2.0 authentication, containerized with Docker, deployed to **AWS EC2**, and integrated as a gRPC client for internal **microservices**; handled production traffic with high reliability.
- Led the **AWS** SDK for JavaScript v2 to v3 migration across 180+ call sites and 11+ **AWS** services (**Lambda**, CloudWatch, SecurityHub, X-Ray, **EC2**, and others) in production Node.js **Lambda** functions, reducing bundle size through modular imports and keeping the platform supported past EOL.
- Designed multi-distro Linux smoke tests (CentOS, Ubuntu ARM64, macOS) for data CLI tools in GitHub Actions release pipelines, hardening deployment safety and catching regressions; unified release workflows and automated Slack notifications for four open-source products.
- Conducted root cause analysis on production incidents — diagnosed a **Go** service stuck in Kubernetes CrashLoopBackOff by investigating pod logs and service status with the platform team — and implemented preventive measures to avoid recurrence.

Kredey (Kredey Innovations Pvt. Ltd.) Remote

2021

Full-Stack Website Developer (Intern)

- Built **full-stack React** + Firebase PWA with full-text and semantic search powered by Elasticsearch, supporting fast data discovery over a Firebase NoSQL backend; implemented real-time notifications and responsive UI/UX.

PROJECTS

TODO / Tasks REST API *(April 2026)*

Full-stack REST API in **TypeScript** with Bun, Express.js 5, and SQLite (bun:sqlite). JWT authentication (access, ID, refresh tokens), RBAC, prepared-statement CRUD service layer with proper REST status codes, database migrations, and route observability middleware. Demonstrates **full-stack** development with security patterns, schema validation (Zod), and production API design.

Lila Tic-Tac-Toe — Real-Time Multiplayer Game *(2026)*

Full-stack real-time system: **React 19** + **TypeScript** frontend on Vercel; **Go**-based Nakama game server with **PostgreSQL** on Fly.io. Server-authoritative architecture over WebSocket transport, KV-storage matchmaking, optimistic UI with failure recovery, reconnect-into-active-match reliability. Demonstrates **distributed systems**, real-time communication, database transactions, and multi-tier deployment.

EDUCATION

University of Engineering & Management (UEM) Kolkata, West Bengal, India

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design